

PROGRAMMING FUNDAMENTAL THEORY

Programming Fundamental Project Programs

PROGRAM NO:1

Electricity Bill Management System:

```
#include <iostream> using
namespace std;

string customername;
int customertype; int
customerID; int
metertype;

void registercustomer()
{
    cout<<"enter customer name:"<<endl;
    cin>>customername;    cout<<"enter
customer ID"<<endl;    cin>>customerID;
```

PROGRAMMING FUNDAMENTAL THEORY

```
    cout<<"1. household\n 2. commercial\n"<<endl;

cout<<"customertype:"<<endl; cin>>customertype;

cout<<"1. Firstmeter\n 2. Secondmeter\n"<<endl;

cout<<"enter meter type"<<endl;      cin>>metertype;

}

double calculateconsumption(int units)
{
    double rates[8]={12.21,14.53,31.51,38.41,41.62,43.04,44.18,49.10};

    if(units<=100) return
units*rates[0]; else
if(units<=200)      return
units*rates[1];

    else if(units<=300)
return units*rates[2]; else
if(units<=400)      return
units*rates[3]; else
if(units<=500)      return
units*rates[4];

    else if(units<=600)
        return units*rates[5];

else if(units<=700)
return units*rates[6];

    else
        return units*rates[7];

}

double calculateGst(double amount){

    return amount*0.18;

}
```

PROGRAMMING FUNDAMENTAL THEORY

```
double calculateincometax(double amount){
```

```
    if(customertype==1)
```

```
return amount*0.10;
```

```
    else
```

```
        return amount*0.15;
```

```
}
```

```
double calculateDuty(double amount)
```

```
{
```

```
    return amount * 0.015;
```

```
}
```

```
double calculateFixedCharges(int units)
```

```
{
```

```
    if(units <= 300)
```

```
return 0;
```

```
    else if(units <= 400)
```

```
return 200;
```

```
    else if(units <= 500)
```

```
return 400;
```

```
    else if(units <= 600)
```

```
return 600;
```

```
    else if(units <= 700)
```

```
return 800;
```

```
    else
```

PROGRAMMING FUNDAMENTAL THEORY

```
        return 1000;
    }
    double calculateConnectionCharges()
    {
        double meterCost = 10000;
        double cableCost = 5000;    double
        securityCost = 15000;

        double additionalCharges;

        if(customertype == 1)
        {
            if(metertype == 1)
            additionalCharges = 2500;
            else
                additionalCharges = 5000;
        }
        else
        {
            if(metertype == 1)
            additionalCharges = 35000;
            else
                additionalCharges = 70000;
        }

        double newConnectionCost = 250000;

        return meterCost + cableCost +securityCost +additionalCharges +newConnectionCost;
```

PROGRAMMING FUNDAMENTAL THEORY

```
}  
  
void displayCustomerDetails()  
{  
    cout << "\n===== CUSTOMER DETAILS =====\n";  
    cout << "Name : " << customername << endl;    cout  
<< "ID  : " << customerID << endl;  
  
    if (customertype == 1)    cout <<  
"Type : Household" << endl;  
    else    cout << "Type : Commercial"  
<< endl;  
}  
  
void displayBill()  
{    int  
units;  
  
    cout << "Enter Units Consumed: ";  
    cin >> units;  
  
    double consumption = calculateconsumption(units);  
    double duty = calculateDuty(consumption);    double  
fixedCharges = calculateFixedCharges(units);    double  
gst = calculateGst(consumption);    double incomeTax =  
calculateincometax(consumption);  
  
    double meterRent = 250;  
    double tvFee = 35;
```

PROGRAMMING FUNDAMENTAL THEORY

```
double totalBill =consumption+duty+fixedCharges+meterRent+tvFee+gst+incomeTax;  cout << "Total
Bill = " << totalBill;

}

int main()

{

    registercustomer();


    int choice;


    do

    {

        cout << "\n 1. Calculate Bill";
cout << "\n 2. New Connection";
cout << "\n 3. Customer Details";
cout << "\n 4. Exit";


        cout << "\nEnter Choice: ";

cin >> choice;


        switch(choice)

        {

            case 1:

displayBill();

break;


            case 2:

                cout << calculateConnectionCharges();

                break;
```

PROGRAMMING FUNDAMENTAL THEORY

case 3:

```
displayCustomerDetails();
```

```
break;
```

case 4:

```
cout << "Program Closed";
```

```
break;
```

default:

```
cout << "Invalid Choice";
```

```
}
```

```
} while(choice != 4);
```

```
return 0;
```

```
}
```

PROGRAMMING FUNDAMENTAL THEORY

```
C:\Users\DELL\OneDrive\Docu  X  +  v

enter customer name:
Usman
enter customer ID
333
1. household
2. commercial

customertype:
2
1. Firstmeter
2. Secondmeter

enter meter type
2

1. Calculate Bill
2. New Connection
3. Customer Details
4. Exit
Enter Choice: 1
Enter Units Consumed: 3
Total Bill = 334.267
1. Calculate Bill
2. New Connection
3. Customer Details
4. Exit
Enter Choice: 1
Enter Units Consumed:
2
Total Bill = 317.845
1. Calculate Bill
2. New Connection
3. Customer Details
4. Exit
Enter Choice: 4
Program Closed

-----
Process exited after 36.88 seconds with return value 0
Press any key to continue . . . |
```

PROGRAM NO:2

Restaurant Management System:

PROGRAMMING FUNDAMENTAL THEORY

```
#include <iostream>

using namespace std;

string customerName;

string
contactNumber; int
orderType; int
persons;
```

```
string items[8] = {
    "Chicken Burger",
    "Zinger Burger",
    "Pizza Small",
    "Pizza Large",
    "Chicken Biryani",
    "BBQ Platter",
    "Fries",
    "Cold Drink"
};
```

```
double prices[8] = {
    450, 550, 900, 1800,
    350, 1200, 250, 120
};
```

PROGRAMMING FUNDAMENTAL THEORY

```
double foodBill = 0;
```

```
void registerCustomer()
```

```
{
```

```
    cout << "Enter Customer Name: ";
```

```
    cin >> customerName;
```

```
    cout << "Enter Contact Number: ";
```

```
    cin >> contactNumber;
```

```
    cout << "1. Dine-in\n2. Takeaway\n";
```

```
    cout << "Enter Order Type: ";    cin >>
```

```
    orderType;
```

```
    cout << "Enter Number of Persons: ";
```

```
    cin >> persons;
```

```
}
```

```
void displayFoodMenu()
```

```
{
```

```
    cout << "\n===== FOOD MENU =====\n";
```

```
    for(int i=0; i<8; i++)
```

```
{
```

PROGRAMMING FUNDAMENTAL THEORY

```
        cout << i+1 << ". "
    << items[i]
        << " - Rs. "
        << prices[i]
        << endl;
    }
}

void placeOrder()
{
    int itemNo, quantity;

    displayFoodMenu();

    cout << "\nEnter Item Number: ";
    cin >> itemNo;

    cout << "Enter Quantity: ";
    cin >> quantity;

    foodBill += prices[itemNo-1] * quantity;

    cout << "Item Added Successfully!\n";
}
```

PROGRAMMING FUNDAMENTAL THEORY

```
double calculateFoodBill()
{
    return
    foodBill;
}

double calculateServiceCharges(double bill)
{
    if(orderType ==
    1)        return bill *
    0.10;    else
    return bill * 0.05;
}

double calculateGST(double bill)
{
    return bill *
    0.16;
}

double calculateDiscount(double bill)
{
    if(bill >= 3000 && bill <= 5000)
    return bill * 0.05;

    else if(bill > 5000 && bill <= 10000)
        return bill * 0.10;
```

PROGRAMMING FUNDAMENTAL THEORY

```
    else if(bill > 10000)
return bill * 0.15;
    else
return 0;
}
```

```
void freeDelivery(double totalBill)
{
    if(totalBill > 5000)
        cout << "Free Delivery Available!\n";
}
```

```
void displayCustomerDetails()
{
    cout << "\n===== CUSTOMER DETAILS =====\n";
    cout << "Name: " << customerName << endl;    cout <<
    "Contact: " << contactNumber << endl;
```

```
    if(orderType == 1)
        cout << "Order Type: Dine-in\n";
    else
        cout << "Order Type: Takeaway\n";
```

PROGRAMMING FUNDAMENTAL THEORY

```
    cout << "Persons: " << persons << endl;
}
```

```
void displayFinalBill()
{
    double serviceCharges =
calculateServiceCharges(foodBill);
```

```
    double gst =
calculateGST(foodBill);
```

```
    double discount =
calculateDiscount(foodBill);
```

```
    double total =
foodBill +
serviceCharges +
    gst -
```

```
discount;    cout
<<
```

```
"\n=====
```

```
RESTAURANT
```

PROGRAMMING FUNDAMENTAL THEORY

BILL

```
=====\\n";
```

```
cout << "Customer Name: "
```

```
<< customerName << endl;
```

```
if(orderType == 1)
```

```
    cout << "Order Type: Dine-in\\n";
```

```
else
```

```
    cout << "Order Type: Takeaway\\n";
```

```
cout << "Number of Persons: "
```

```
<< persons << endl;
```

```
cout << "\\nFood Bill: Rs. "
```

```
<< foodBill << endl;
```

```
cout << "Service Charges: Rs. "
```

```
<< serviceCharges << endl;
```

```
cout << "GST: Rs. "
```

```
<< gst << endl;
```

PROGRAMMING FUNDAMENTAL THEORY

```
cout << "Discount: Rs. "
```

```
<< discount << endl;
```

```
cout << "-----\n";
```

```
cout << "Total Payable Amount: Rs. "
```

```
<< total << endl;
```

```
freeDelivery(total);
```

```
cout << "Enjoy Your Meal :)\n";
```

```
cout << "=====\n";
```

```
}
```

```
int main() {
```

```
    registerCustomer();
```

```
    int choice;
```

```
    do
```

```
    {
```

```
        cout << "\n===== MAIN MENU =====\n";
```


PROGRAMMING FUNDAMENTAL THEORY

```
cout << "1. View Food Menu\n";  
cout << "2. Place Order\n";    cout <<  
"3. Calculate Bill\n";    cout << "4. View  
Customer Details\n";    cout << "5.  
Exit\n";
```

```
    cout << "Enter Choice: ";  
    cin >> choice;
```

```
        switch(choice)  
        {  
case 1:  
            displayFoodMenu();  
break;  
        case 2:  
            placeOrder();  
break;  
        case 3:  
            displayFinalBill();  
break;  
        case 4:  
            displayCustomerDetails();  
break;
```

PROGRAMMING FUNDAMENTAL THEORY

case

5:

cout << "Program Closed.\n";

break;

default:

cout << "Invalid Choice!\n";

}

} while(choice != 5);

return 0;

}

PROGRAMMING FUNDAMENTAL THEORY

```
1. Dine-in
2. Takeaway
Enter Order Type: 2
Enter Number of Persons: 5
```

```
===== MAIN MENU =====
```

```
1. View Food Menu
2. Place Order
3. Calculate Bill
4. View Customer Details
5. Exit
Enter Choice: 2
```

```
===== FOOD MENU =====
```

```
1. Chicken Burger - Rs. 450
2. Zinger Burger - Rs. 550
3. Pizza Small - Rs. 900
4. Pizza Large - Rs. 1800
5. Chicken Biryani - Rs. 350
6. BBQ Platter - Rs. 1200
7. Fries - Rs. 250
8. Cold Drink - Rs. 120
```

```
Enter Item Number: 4
Enter Quantity: 2
Item Added Successfully!
```

```
===== MAIN MENU =====
```

```
1. View Food Menu
2. Place Order
3. Calculate Bill
4. View Customer Details
5. Exit
Enter Choice: 1
```

```
===== FOOD MENU =====
```

```
1. Chicken Burger - Rs. 450
2. Zinger Burger - Rs. 550
3. Pizza Small - Rs. 900
4. Pizza Large - Rs. 1800
5. Chicken Biryani - Rs. 350
6. BBQ Platter - Rs. 1200
7. Fries - Rs. 250
8. Cold Drink - Rs. 120
```

```
===== MAIN MENU =====
```

```
1. View Food Menu
2. Place Order
3. Calculate Bill
4. View Customer Details
5. Exit
Enter Choice: 3
```

```
===== RESTAURANT BILL =====
```

```
Customer Name: usman
Order Type: Takeaway
Number of Persons: 5
```

```
Food Bill: Rs. 3600
Service Charges: Rs. 180
GST: Rs. 576
Discount: Rs. 180
```

```
-----
Total Payable Amount: Rs. 4176
Enjoy Your Meal :)
```

```
=====
```

```
===== MAIN MENU =====
```

```
1. View Food Menu
2. Place Order
3. Calculate Bill
4. View Customer Details
5. Exit
Enter Choice: 5
Program Closed.
```

```
-----
Process exited after 52.94 seconds with return value 0
```

PROGRAMMING FUNDAMENTAL THEORY

Program No: 3

Grocery Mart Management System:

```
#include <iostream>

using namespace std;

string customerName;
int customerID; int
customerType; int
paymentMethod;

double grossBill = 0;
string items[8] = {
    "Rice 1 KG",
        "Sugar 1 KG",
        "Cooking Oil 1 Litre",
        "Milk Pack",
        "Tea Pack",
        "Flour 5 KG",
        "Eggs Dozen",
        "Detergent"
};
```

PROGRAMMING FUNDAMENTAL THEORY

```
double prices[8] = {  
    350,180,580,220,  
    450,950,320,600  
};
```

```
void registerCustomer()  
{  
    cout << "Enter Customer Name: ";  
    cin >> customerName;  
  
    cout << "Enter Customer ID: ";  
    cin >> customerID;  
  
    cout << "\n1. Regular Customer";  
    cout << "\n2. Member Customer";  
    cout << "\nEnter Customer Type: ";  
    cin >> customerType;  
  
    cout << "\n1. Cash";    cout << "\n2.  
Card";    cout << "\nEnter Payment  
Method: ";    cin >> paymentMethod;  
}
```

PROGRAMMING FUNDAMENTAL THEORY

```
void displayGroceryList()
{
    cout << "\n===== GROCERY ITEMS =====\n";

    for(int i=0; i<8; i++)
    {
        cout << i+1 << ". "
<< items[i]
        << " - Rs. "
        << prices[i] << endl;
    }
}
```

```
void addItemToCart()
{
    int itemNo, quantity;

    displayGroceryList();

    cout << "\nEnter Item Number: ";
    cin >> itemNo;

    cout << "Enter Quantity: ";
    cin >> quantity;
```

PROGRAMMING FUNDAMENTAL THEORY

```
grossBill += prices[itemNo-1] * quantity;
```

```
cout << "Item Added Successfully!\n";
```

```
}
```

```
double calculateGrossBill()
```

```
{    return
```

```
grossBill;
```

```
}
```

```
double calculateSalesTax(double bill)
```

```
{
```

```
    return bill * 0.05;
```

```
}
```

```
double calculateMembershipDiscount(double bill)
```

```
{
```

```
    if(customerType == 2)
```

```
        return bill * 0.07;
```

```
else
```

```
    return 0;
```

```
}
```

PROGRAMMING FUNDAMENTAL THEORY

```
double calculateBillDiscount(double bill)
```

```
{
```

```
    if(bill >= 5000 && bill <= 10000)
```

```
        return bill * 0.05;
```

```
    else if(bill > 10000)
```

```
        return bill * 0.10;
```

```
    else
```

```
        return 0;
```

```
}
```

```
double calculateCardCharges(double bill)
```

```
{
```

```
    if(paymentMethod == 2)
```

```
        return bill * 0.02;
```

```
    else    return 0;
```

```
}
```

```
int calculateLoyaltyPoints(double total)
```

```
{
```

```
    return total / 100;
```

```
}
```


PROGRAMMING FUNDAMENTAL THEORY

```
void redeemPoints(double &total)
{
    int existingPoints;
    int choice;

    int generatedPoints =
    calculateLoyaltyPoints(total);

    cout << "\nGenerated Loyalty Points: "
         << generatedPoints;

    cout << "\nEnter Your Existing Loyalty Points: ";
    cin >> existingPoints;

    int updatedPoints =
    generatedPoints + existingPoints;

    cout << "\nLoyalty Points After Purchase: "
         << updatedPoints;

    cout << "\nPress 1 to Redeem Points";
    cout << "\nPress 2 to Continue";    cout
    << "\nEnter Choice: ";    cin >> choice;
```

PROGRAMMING FUNDAMENTAL THEORY

```
if(choice == 1)
{
    total = total - updatedPoints;

    cout << "\nPoints Redeemed Successfully!";
}
}

void displayFinalBill()
{
    double tax
=
    calculateSalesTax(grossBill);

    double memberDiscount =
calculateMembershipDiscount(grossBill);    double billDiscount =
    calculateBillDiscount(grossBill);

    double cardCharges =
    calculateCardCharges(grossBill);

    double total =
grossBill +    tax +
```

PROGRAMMING FUNDAMENTAL THEORY

cardCharges -

memberDiscount -

billDiscount;

```
cout << "\n===== GROCERY MART BILL =====\n";
```

```
cout << "Customer Name: "
```

```
<< customerName << endl;
```

```
if(customerType == 1)
```

```
    cout << "Customer Type: Regular\n";
```

```
else
```

```
    cout << "Customer Type: Member\n";
```

```
if(paymentMethod == 1)    cout <<
```

```
"Payment Method: Cash\n";
```

```
else
```

```
    cout << "Payment Method: Card\n";
```

```
cout << "\nGross Bill: Rs. "
```

```
<< grossBill << endl;
```

```
cout << "Sales Tax: Rs. "
```

```
<< tax << endl;
```

PROGRAMMING FUNDAMENTAL THEORY

```
cout << "Membership Discount: Rs. "  
    << memberDiscount << endl;
```

```
cout << "Bill Discount: Rs. "  
    << billDiscount << endl;
```

```
cout << "Card Charges: Rs. "  
    << cardCharges << endl;
```

```
redeemPoints(total);
```

```
cout << "\n-----\n";    cout << "Total  
Payable Amount: Rs. "  
    << total << endl;
```

```
cout << "Thank You For Shopping :)\n";
```

```
cout << "=====\n";  
}
```

```
void displayCustomerDetails()
```

```
{  
    cout << "\n===== CUSTOMER DETAILS =====\n";
```

PROGRAMMING FUNDAMENTAL THEORY

```
cout << "Name: "  
    << customerName << endl;
```

```
cout << "ID: "  
    << customerID << endl;
```

```
if(customerType == 1)  
cout << "Type: Regular\n";  
else  
    cout << "Type: Member\n";
```

```
if(paymentMethod == 1)  
cout << "Payment: Cash\n";  
else  
    cout << "Payment: Card\n";  
}
```

```
int main() {  
    registerCustomer();
```

```
    int choice;
```

PROGRAMMING FUNDAMENTAL THEORY

```
do
{
    cout << "\n\n===== GROCERY MART MENU =====\n";
    cout << "1. View Grocery Items\n";
cout << "2. Add Items To Cart\n";
cout << "3. Calculate Bill\n";    cout <<
"4. View Customer Details\n";    cout <<
"5. Exit\n";

    cout << "Enter Choice: ";
    cin >> choice;

    switch(choice)
    {
        case 1:
displayGroceryList();
break;

        case
2:
            addItemToCart();
break;

        case 3:
displayFinalBill();
break;
```

PROGRAMMING FUNDAMENTAL THEORY

case

4:

displayCustomerDetails();

break;

case

5:

cout << "Program Closed.\n";

break;

default:

cout << "Invalid Choice!\n";

}

} while(choice != 5);

return 0;

}

PROGRAMMING FUNDAMENTAL THEORY

```
Enter Customer Name: usman
Enter Customer ID: 787
```

1. Regular Customer
2. Member Customer

```
Enter Customer Type: 2
```

1. Cash
2. Card

```
Enter Payment Method: 3
```

```
===== GROCERY MART MENU =====
```

1. View Grocery Items
2. Add Items To Cart
3. Calculate Bill
4. View Customer Details
5. Exit

```
Enter Choice: 4
```

```
===== CUSTOMER DETAILS =====
```

```
Name: usman
ID: 787
Type: Member
Payment: Card
```

```
===== GROCERY MART MENU =====
```

1. View Grocery Items
2. Add Items To Cart
3. Calculate Bill
4. View Customer Details
5. Exit

```
Enter Choice: 1
```

```
===== GROCERY ITEMS =====
```

1. Rice 1 KG - Rs. 350
2. Sugar 1 KG - Rs. 180
3. Cooking Oil 1 Litre - Rs. 580
4. Milk Pack - Rs. 220
5. Tea Pack - Rs. 450
6. Flour 5 KG - Rs. 950
7. Eggs Dozen - Rs. 320
8. Detergent - Rs. 600

```
===== GROCERY MART MENU =====
```

1. View Grocery Items
2. Add Items To Cart
3. Calculate Bill
4. View Customer Details
5. Exit

```
Enter Choice: 5
```

```
Program Closed.
```

```
-----
Process exited after 35.86 seconds with return value 0
Press any key to continue . . . |
```


PROGRAMMING FUNDAMENTAL THEORY

Program No:4

Online Shopping System:

```
#include <iostream> using
```

```
namespace std;
```

```
string userName, email, city;
```

```
int customerType;    int
```

```
paymentMethod;
```

```
double productTotal = 0;
```

```
string products[8] = {
```

```
    "T-Shirt","Jeans","Shoes","Watch",
```

```
    "Handbag","Headphones","Mobile Cover","Perfume"
```

```
};
```

```
double prices[8] = {
```

```
    1200,3500,5000,2500,
```

```
    4200,3000,700,2800
```

```
};
```

```
void registerUser()
```

PROGRAMMING FUNDAMENTAL THEORY

```
{  
    cout << "Enter User Name: ";  
    cin >> userName;  
  
    cout << "Enter Email: ";  
    cin >> email;  
  
    cout << "Enter City: ";  
    cin >> city;  
  
    cout << "\n1. New Customer\n2. Returning Customer";  
    cout << "\nEnter Customer Type: ";    cin >>  
    customerType;  
  
    cout << "\n1. Cash on Delivery\n2. Card";  
    cout << "\nEnter Payment Method: ";    cin  
>> paymentMethod;  
}  
  
void displayProducts()  
{  
    cout << "\n===== PRODUCT LIST =====\n";
```

PROGRAMMING FUNDAMENTAL THEORY

```
for(int i=0; i<8; i++)
{
    cout << i+1 << ". "
<< products[i]
    << " - Rs. "
    << prices[i] << endl;
}
}

void addToCart()
{
    int itemNo, quantity;

    displayProducts();

    cout << "\nEnter Product Number: ";
    cin >> itemNo;

    cout << "Enter Quantity: ";
    cin >> quantity;

    productTotal += prices[itemNo-1] * quantity;    cout << "Product
Added!\n";
}
```

PROGRAMMING FUNDAMENTAL THEORY

```
double calculateProductTotal()
{
    return productTotal;
}
```

```
double calculateGST(double total)
{
    return total * 0.17;
}
```

```
double calculateDeliveryCharges()
{
    if(city == "Lahore" || city == "Karachi" || city == "Islamabad")
return 250;    else        return 500;
}
```

```
double calculateCustomerDiscount(double total)
{
    if(customerType == 1)
return total * 0.05;
```

PROGRAMMING FUNDAMENTAL THEORY

```
else    return total *  
0.10;  
}
```

```
double calculateOrderDiscount(double total)  
{  
    if(total >= 5000 && total <= 10000)  
return total * 0.05;    else if(total >  
10000)    return total * 0.12;    else  
return 0;  
}
```

```
double calculatePaymentCharges(double total)  
{  
    if(paymentMethod == 2)  
return total * 0.025;    else  
return 0;  
}
```

```
void displayBill()  
{  
    double gst = calculateGST(productTotal);    double delivery  
= calculateDeliveryCharges();    double custDisc =
```

PROGRAMMING FUNDAMENTAL THEORY

```
calculateCustomerDiscount(productTotal);    double orderDisc  
= calculateOrderDiscount(productTotal);    double payment =  
calculatePaymentCharges(productTotal);
```

```
    double finalTotal =  
productTotal +  
gst +    delivery +  
payment -  
custDisc -  
orderDisc;
```

```
    cout << "\n===== ONLINE SHOPPING BILL  
=====\\n";
```

```
    cout << "User Name: " << userName << endl;  
    cout << "City: " << city << endl;
```

```
    if(customerType == 1)  
        cout << "Customer Type: New Customer\\n";  
    else  
        cout << "Customer Type: Returning Customer\\n";
```

```
    cout << "\nProduct Total: Rs. " << productTotal << endl;  
    cout << "GST: Rs. " << gst << endl;
```

PROGRAMMING FUNDAMENTAL THEORY

```
cout << "Delivery Charges: Rs. " << delivery << endl;
cout << "Customer Discount: Rs. " << custDisc << endl;
cout << "Order Discount: Rs. " << orderDisc << endl;    cout
<< "Payment Charges: Rs. " << payment << endl;

cout << "-----\n";

cout << "Final Payable Amount: Rs. " << finalTotal << endl;

cout << "Thank You for Shopping :)\n";
cout <<
"===== \n";
}

void viewUserDetails()
{
    cout << "\n===== USER DETAILS =====\n";
    cout << "Name: " << userName << endl;    cout
<< "Email: " << email << endl;    cout << "City:
" << city << endl;

    if(customerType == 1)    cout <<
"Type: New Customer\n";    else
```

PROGRAMMING FUNDAMENTAL THEORY

```
    cout << "Type: Returning Customer\n";
}

int main() {
    registerUser();

    int choice;

    do
    {
        cout << "\n===== ONLINE SHOP MENU =====\n";        cout <<
"1. View Products\n";        cout << "2. Add Product to Cart\n";        cout
<< "3. Calculate Checkout Bill\n";        cout << "4. View User Details\n";
cout << "5. Exit\n";

        cout << "Enter Choice: ";
cin >> choice;

        switch(choice)
        {
            case 1:
displayProducts();
break;
```


PROGRAMMING FUNDAMENTAL THEORY

case 2:

addToCart();

break;

case 3:

displayBill();

break;

case 4:

viewUserDetails();

break;

case

5:

cout << "Program Closed.\n";

break;

default:

cout << "Invalid Choice!\n";

}

} while(choice != 5);

return 0;

}

PROGRAMMING FUNDAMENTAL THEORY

```
Enter User Name: usman
Enter Email: usman@gmail.com
Enter City: lahore
```

```
1. New Customer
2. Returning Customer
Enter Customer Type: 1
```

```
1. Cash on Delivery
2. Card
Enter Payment Method: 2
```

```
===== ONLINE SHOP MENU =====
```

```
1. View Products
2. Add Product to Cart
3. Calculate Checkout Bill
4. View User Details
5. Exit
Enter Choice: 1
```

```
===== PRODUCT LIST =====
```

```
1. T-Shirt - Rs. 1200
2. Jeans - Rs. 3500
3. Shoes - Rs. 5000
4. Watch - Rs. 2500
5. Handbag - Rs. 4200
6. Headphones - Rs. 3000
7. Mobile Cover - Rs. 700
8. Perfume - Rs. 2800
```

```
===== ONLINE SHOP MENU =====
```

```
1. View Products
2. Add Product to Cart
3. Calculate Checkout Bill
4. View User Details
5. Exit
Enter Choice: 5
Program Closed.
```

```
-----
Process exited after 36.71 seconds with return value 0
Press any key to continue . . . |
```